

CLASS 9

MATHEMATICS

Topic: Coordinate Geometry

Solved Exercise — Questions 11 to 16

(Ganita Manjari | Grade 9 | CBSE Standard)

Question 11: Let P, Q be points of trisection of AB, with P closer to A and Q closer to B. Find the coordinates of P and Q for A(4, 7) and B(16, -2).

Concept — Section Formula:

If a point divides the line segment joining (x_1, y_1) and (x_2, y_2) in the ratio $m : n$, its coordinates are:

$$x = \frac{mx_2 + nx_1}{m+n}, y = \frac{my_2 + ny_1}{m+n}$$

Trisection: P divides AB in ratio 1:2 and Q divides AB in ratio 2:1.

Finding P (ratio 1:2):

$$x_P = \frac{1 \times 16 + 2 \times 4}{1+2} = \frac{16+8}{3} = \frac{24}{3} = 8$$

$$y_P = \frac{1 \times (-2) + 2 \times 7}{1+2} = \frac{-2+14}{3} = \frac{12}{3} = 4$$

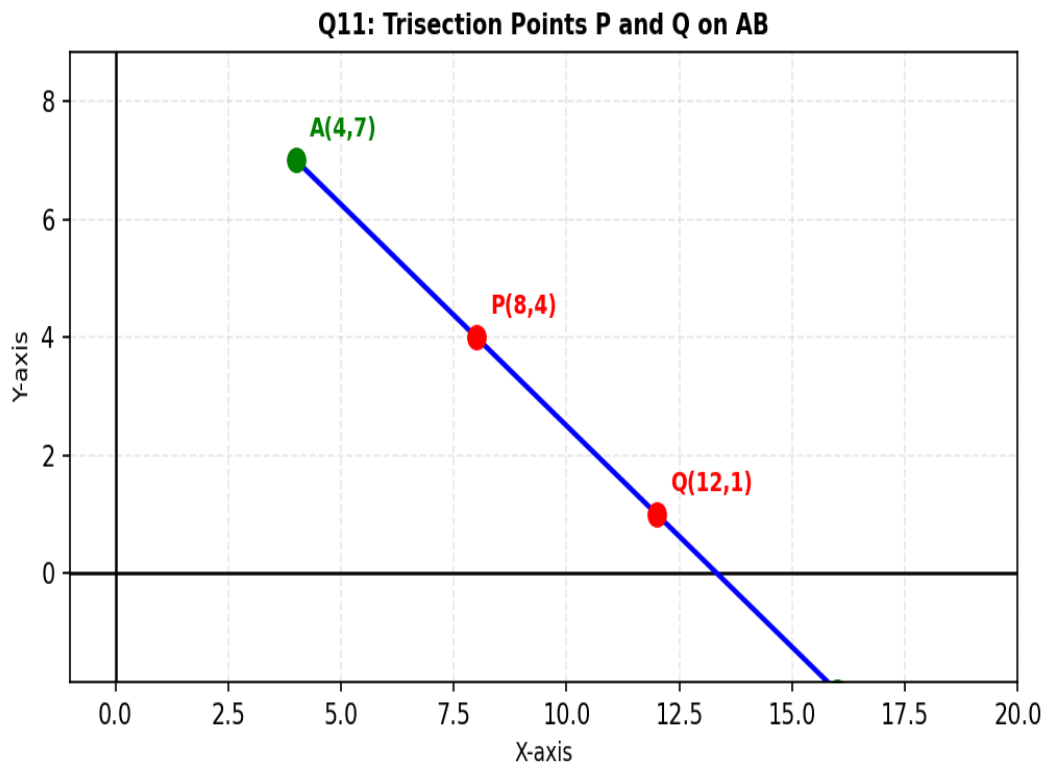
$$\therefore P = (8, 4)$$

Finding Q (ratio 2:1):

$$x_Q = \frac{2 \times 16 + 1 \times 4}{2+1} = \frac{32+4}{3} = \frac{36}{3} = 12$$

$$y_Q = \frac{2 \times (-2) + 1 \times 7}{2+1} = \frac{-4+7}{3} = \frac{3}{3} = 1$$

$$\therefore Q = (12, 1)$$



✓ Verification: Midpoint of PQ = $\left(\frac{8+12}{2}, \frac{4+1}{2}\right) = (10, 2.5)$ = Midpoint of AB. ✓

Question 12(i): Given A(1, -8), B(-4, 7) and C(-7, -4), show they lie on circle K centred at origin O(0, 0). Find the radius.

A point lies on a circle if its distance from the centre equals the radius.

Distance formula: $d = \sqrt{(x-h)^2 + (y-k)^2}$, where centre (h, k) = (0, 0).

$$OA = \sqrt{1^2 + (-8)^2} = \sqrt{1 + 64} = \sqrt{65}$$

$$OB = \sqrt{(-4)^2 + 7^2} = \sqrt{16 + 49} = \sqrt{65}$$

$$OC = \sqrt{(-7)^2 + (-4)^2} = \sqrt{49 + 16} = \sqrt{65}$$

Since $OA = OB = OC = \sqrt{65}$, all three points are equidistant from O.

$\therefore$ A, B, C lie on circle K with radius $r = \sqrt{65} \approx 8.06$ units.

Question 12(ii): Check whether D(-5, 6) and E(0, 9) lie within, on, or outside circle K.

Compare distance from O with $r = \sqrt{65}$.

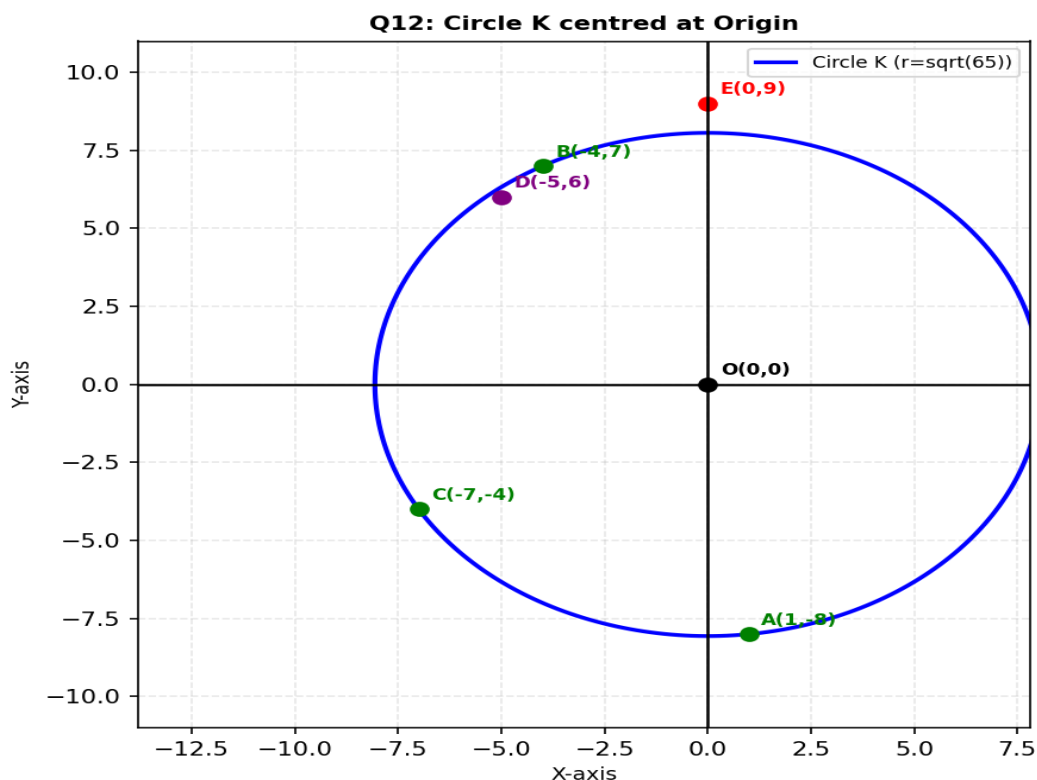
$$OD = \sqrt{(-5)^2 + 6^2} = \sqrt{25 + 36} = \sqrt{61}$$

$\sqrt{61} < \sqrt{65} \rightarrow$ D is **inside** circle K.

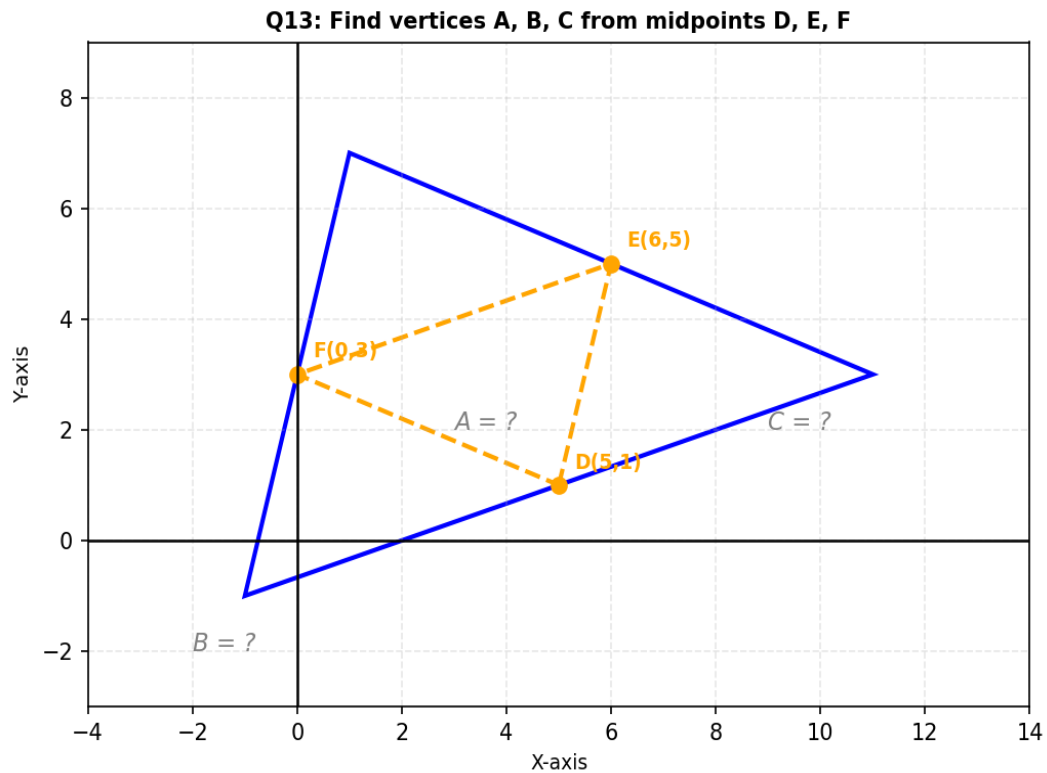
$$OE = \sqrt{0^2 + 9^2} = \sqrt{81} = 9$$

$9^2 = 81 > 65 = r^2 \rightarrow$ E is **outside** circle K.

$\therefore$ D lies **INSIDE** circle K; E lies **OUTSIDE** circle K.



Question 13: The midpoints of the sides of triangle ABC are D(5, 1), E(6, 5) and F(0, 3), respectively. Find the coordinates of A, B and C.



■ HOMEWORK

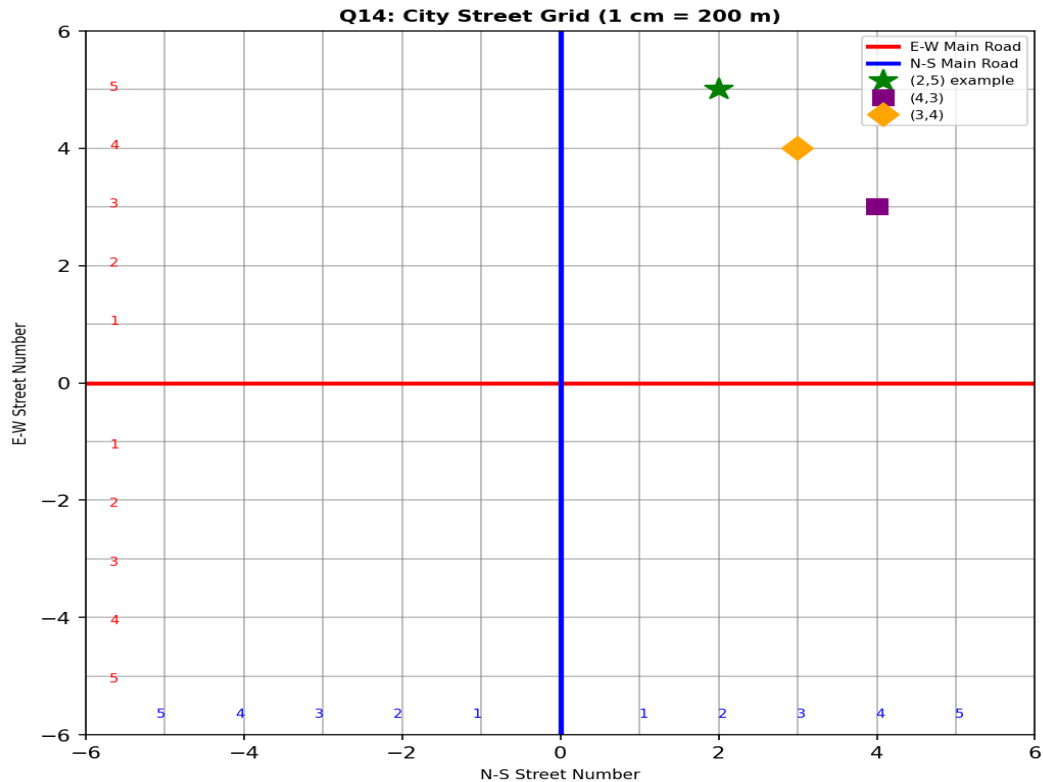
This question is assigned as **Homework**.

Use the midpoint formula and simultaneous equations to find the vertices A, B and C of the triangle.

Hint: If D, E, F are midpoints of BC, AC, AB respectively, then set up three pairs of equations and solve for x and y coordinates.

Question 14: A city has two main roads crossing at the centre along N-S and E-W directions. Streets are 200 m apart; 10 streets each direction. Using 1 cm = 200 m, draw the model and find:

(i) The city grid model is drawn below. Origin = city centre. Streets numbered 1-5 on each side of both main roads.



(ii) **Convention:** Intersection (m, n) = m-th N-S street × n-th E-W street.

Example: (2, 5) = 2nd N-S street meeting 5th E-W street.

(a) How many intersections can be called (4, 3)?

There are 2 streets labelled "4" in N-S direction (one east, one west of main road) and 2 streets labelled "3" in E-W direction (one north, one south). Total = $2 \times 2 = 4$.

∴ (4, 3) refers to 4 different street intersections.

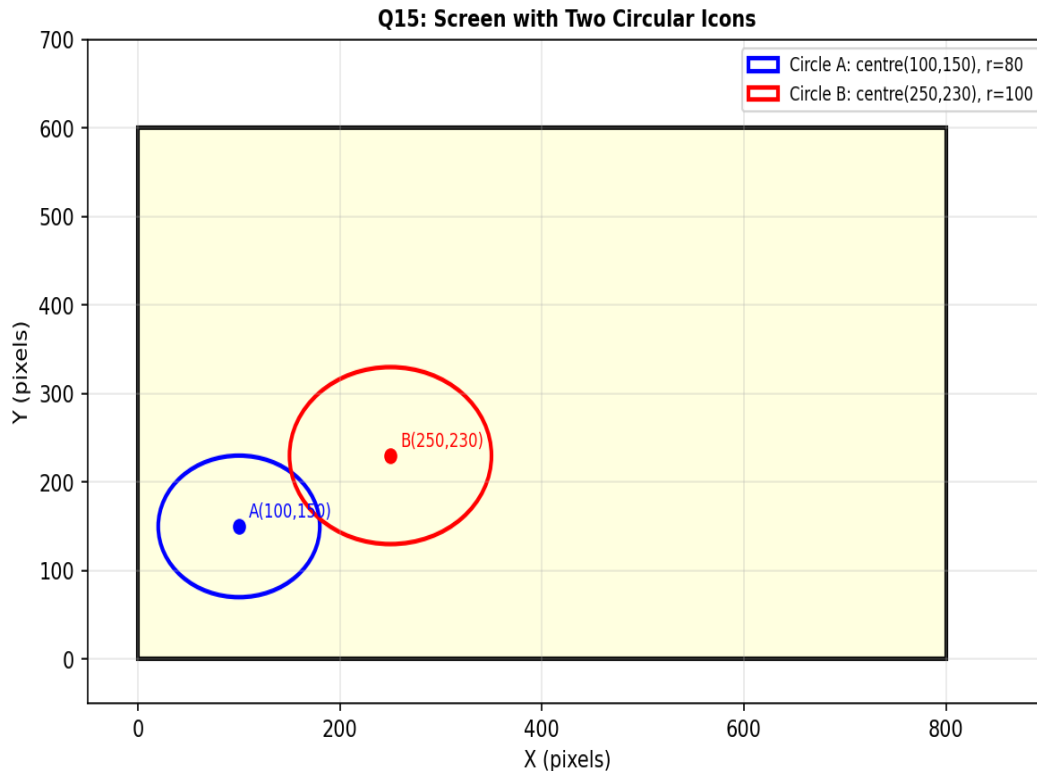
(b) How many intersections can be called (3, 4)?

Similarly, 2 N-S streets numbered "3" and 2 E-W streets numbered "4". Total = $2 \times 2 = 4$.

∴ (3, 4) also refers to 4 different street intersections.

Note: If directional labels (N/S/E/W) are used, each gives exactly 1 intersection.

Question 15: A screen is 800×600 pixels. Circle A: centre (100, 150), r = 80 px. Circle B: centre (250, 230), r = 100 px. (i) Does any part lie outside the screen? (ii) Do the circles intersect?



(i) A circle stays within the screen if:

$cx - r \geq 0$, $cx + r \leq W$, $cy - r \geq 0$, $cy + r \leq H$

Circle A [cx=100, cy=150, r=80]:

Left: $100 - 80 = 20 \geq 0$ ✓ Right: $100 + 80 = 180 \leq 800$ ✓ Bottom: $150 - 80 = 70 \geq 0$ ✓ Top: $150 + 80 = 230 \leq 600$ ✓

Circle A lies entirely within the screen.

Circle B [cx=250, cy=230, r=100]:

Left: $250 - 100 = 150 \geq 0$ ✓ Right: $250 + 100 = 350 \leq 800$ ✓ Bottom: $230 - 100 = 130 \geq 0$ ✓ Top: $230 + 100 = 330 \leq 600$ ✓

Circle B also lies entirely within the screen. Neither circle extends outside.

(ii) Do the circles intersect?

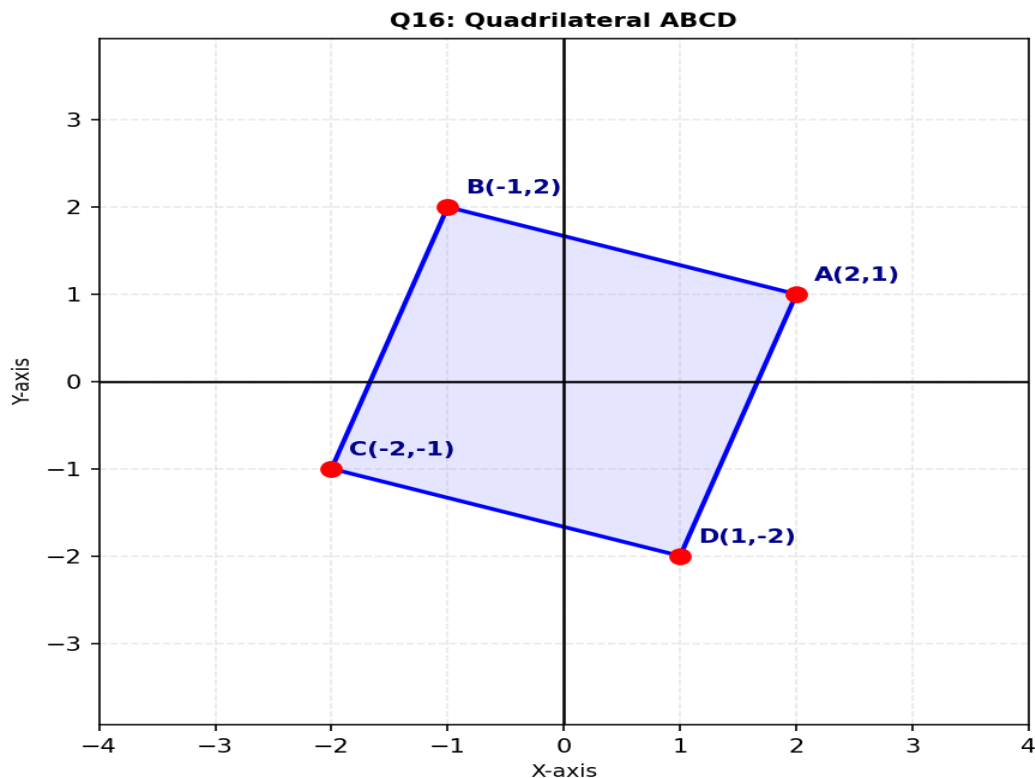
$$d = \sqrt{(250 - 100)^2 + (230 - 150)^2} = \sqrt{150^2 + 80^2} = \sqrt{28900} = 170$$

Sum of radii = $80 + 100 = 180$ | Difference of radii = $|100 - 80| = 20$

Since $20 < 170 < 180 \rightarrow |r_1 - r_2| < d < r_1 + r_2$

∴ The two circles INTERSECT each other at two points.

Question 16: Plot A(2, 1), B(-1, 2), C(-2, -1) and D(1, -2) in the coordinate plane. Is ABCD a square? Find its area.



Step 1 — Calculate all four sides:

$$AB = \sqrt{[2 - (-1)]^2 + (1 - 2)^2} = \sqrt{[9 + 1]} = \sqrt{10}$$

$$BC = \sqrt{[-1 - (-2)]^2 + (2 - (-1))^2} = \sqrt{[1 + 9]} = \sqrt{10}$$

$$CD = \sqrt{[-2 - 1]^2 + (-1 - (-2))^2} = \sqrt{[9 + 1]} = \sqrt{10}$$

$$DA = \sqrt{[1 - 2]^2 + (-2 - 1)^2} = \sqrt{[1 + 9]} = \sqrt{10}$$

All four sides equal: $AB = BC = CD = DA = \sqrt{10}$ ✓

Step 2 — Check diagonals:

$$AC = \sqrt{[2 - (-2)]^2 + (1 - (-1))^2} = \sqrt{[16 + 4]} = 2 \cdot \sqrt{5}$$

$$BD = \sqrt{[-1 - 1]^2 + (2 - (-2))^2} = \sqrt{[4 + 16]} = 2 \cdot \sqrt{5}$$

Both diagonals equal: $AC = BD = 2 \cdot \sqrt{5}$ ✓

Step 3 — Perpendicularity of diagonals:

$$\text{Slope of AC} = \frac{(-1 - 1)}{(-2 - 2)} = \frac{-2}{-4} = \frac{1}{2}$$

$$\text{Slope of BD} = \frac{(-2 - 2)}{(1 - (-1))} = \frac{-4}{2} = -2$$

$$\text{Product} = \left(\frac{1}{2}\right) \times (-2) = -1 \rightarrow AC \perp BD \quad \checkmark$$

All sides equal + equal perpendicular diagonals $\rightarrow$ ABCD is a **SQUARE**.

Step 4 — Area:

$$\text{Method 1 (side): Area} = (\sqrt{10})^2 = \mathbf{10 \text{ sq. units}}$$

$$\text{Method 2 (diagonals): Area} = \frac{d^2}{2} = \frac{(2 \cdot \sqrt{5})^2}{2} = \frac{20}{2} = \mathbf{10 \text{ sq. units}} \quad \checkmark$$

$\therefore$ ABCD is a SQUARE with area = 10 square units.